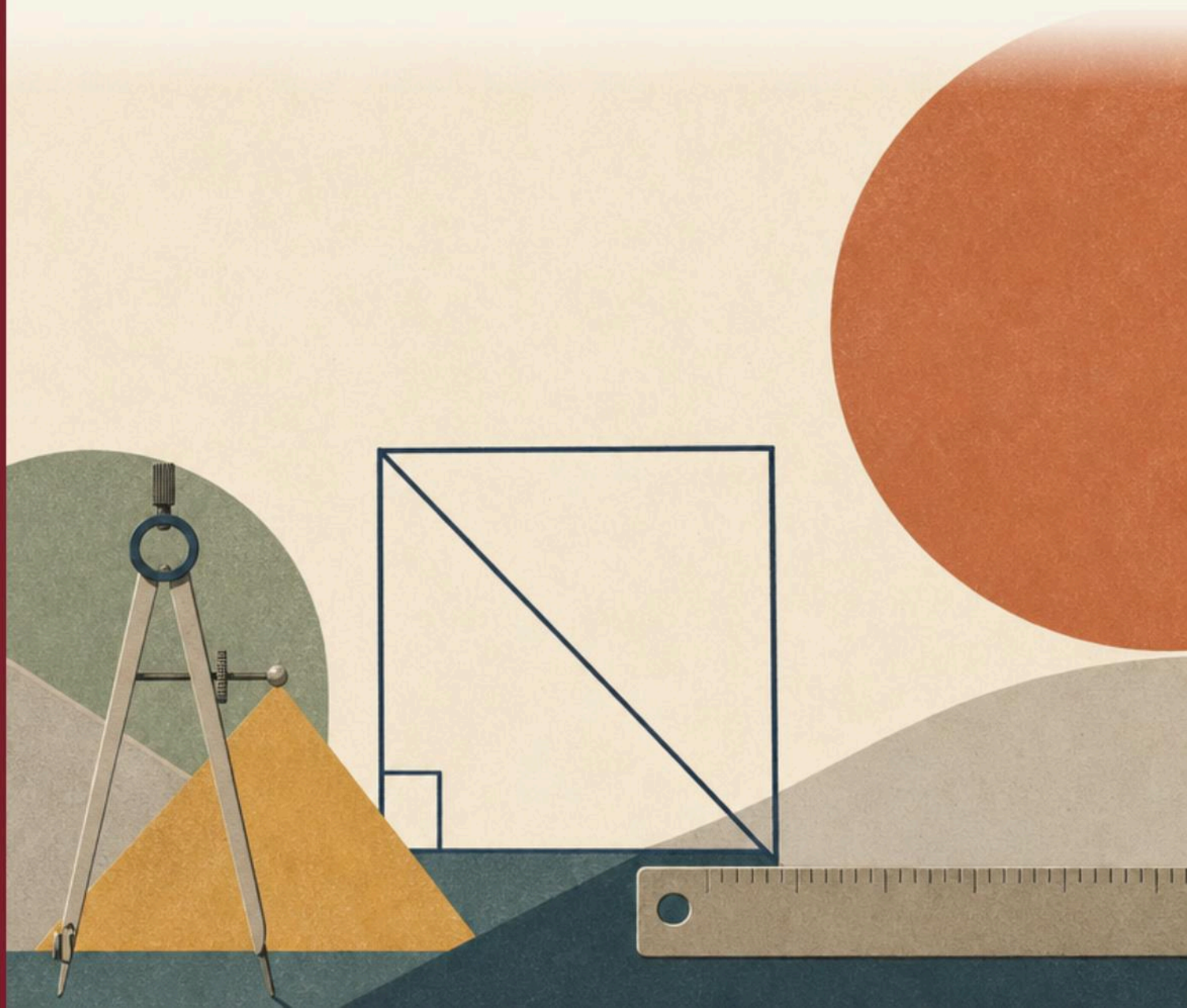


Mathematics

Year 9

Cambridge Lower Secondary

Student Manual



Prime Books • Mathematics Year 9 • Student Book

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Every scenario, dataset, name and worked example in this book is original to this edition. Factual and historical references are drawn from real, credited sources, which are listed in the Sources & references section at the back of this book.

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Printed for Prime School • Year 9 • Mathematics

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INTRODUCTION

How this book works

Welcome to Prime Books Mathematics for Year 9. This book is written for you, the pupil, to read, argue with and write in. It is not a book to be afraid of. Every idea is introduced with a reason before it becomes a rule, and every rule is followed by the chance to use it until it feels obvious.

Mathematics at this level is about making the familiar strange and the strange familiar. You already know how to add, subtract, multiply and divide. In this first unit you will stretch those four operations across the whole number line, into the territory of the negatives, and then use them to unlock older and deeper patterns: multiples, factors, quick tests for divisibility, and the roots that undo powers.

What makes this book different

- **Every concept has a reason.** You will rarely meet a bare rule. You will meet a situation that forces the rule into being.
- **Every example is worked in full.** Nothing is left as "an exercise for the reader" in the middle of a new idea.
- **You are asked to think like a mathematician.** Several times in each unit you are invited to investigate, conjecture and justify, not just calculate.
- **Everything is real.** Names, places and stories are drawn from life in

Portugal, and every fact can be checked in the Sources & references at the back.

A note to teachers and parents

This book follows the spirit and sequence of an international lower-secondary mathematics programme, interpreted afresh for Prime School. Each unit opens with diagnostic questions to surface prior knowledge, builds concepts through worked examples and structured practice, and closes with self-assessment and a summative check. The pedagogy is deliberately investigation-led: pupils are asked to notice patterns before being handed formulas, because mathematics that is discovered is mathematics that is remembered.

HOW TO USE THIS BOOK

Reading the page

Each unit follows the same rhythm, so you always know where you are. Here is what each kind of panel is for.

PANEL	WHAT IT DOES
Get started	A few quick questions to wake up what you already know. No new ideas here.
Scenario	A real situation that makes the unit's maths worth learning.
Learning outcomes	The promises this unit makes to you. Tick them off as you master them.
Warm up	A short puzzle to get your mathematical blood moving.
Do you remember?	A gentle reminder of earlier work the new idea depends on.
Learn	The main teaching. Read slowly; this is where new ideas are born.
Worked example	A problem solved in full, with the reasoning shown beside it.
Practise	Questions for you. Fluency first, then reasoning, then a stretch.
Did you know?	A true and surprising fact connected to the topic.
Think like a mathematician	An open investigation. Conjecture, test, justify.
Go further	For when you want more: a harder idea or a wider view.
Challenge yourself	The stretch. Attempt these; they are meant to be hard.
Final project	A bigger task that pulls the whole unit together.
Check your progress	A mixed set of questions across the whole unit.
Evaluation	Be honest with yourself: green, amber or red on each outcome.
What can you do?	A checklist to confirm what you can now do.
Keywords	The mathematical words this unit has taught you.

How to work

Read with a pencil in your hand. When the book asks a question, stop and try it before you read the answer. Mathematics is not a spectator sport: you learn it by doing it, getting it wrong, and seeing why.

GETTING SET UP

Before Unit 1

You do not need much for this unit: a pencil, a ruler, a rubber, and a squared exercise book. A calculator is useful for checking, but most of this unit is about doing without one, so keep it in your bag until you are asked.

Your mathematical toolkit for this unit

- A **number line** you can sketch quickly, with zero in the middle.
- Times tables up to 12×12 , fluent enough that you do not have to think.
- The four operations on positive whole numbers, secure.
- Curiosity. This is the most important tool.

A quick self-check

Try these now. If any feel hard, tell your teacher before you begin: the unit builds on them.

- 1 Write the numbers -3 , 5 , 0 , -7 , 2 in order, smallest first.
- 2 List the first four multiples of 7 .
- 3 List all the factors of 20 .
- 4 Work out 13×4 and $96 \div 8$ without a calculator.

The answers, and the ideas behind them, are exactly what Unit 1 will sharpen. When you can do these confidently, you are ready.

UNIT 1

1

Number and calculation

Irrational numbers, standard form and indices

© LEARNING OUTCOMES

- ✓ Distinguish rational and irrational numbers.
- ✓ Write and calculate with numbers in standard form.
- ✓ Use zero, negative and fractional indices.
- ✓ Estimate with roots and powers.

◆ GET STARTED

- 1 Write 84 as a product of prime factors.
- 2 Work out: a. 3^4 b. 2^6 c. $(-2)^3$.
- 3 Work out, giving your answer as a power: a. $5^3 \times 5^2$ b. $7^8 \div 7^5$.
- 4 Is $\sqrt{25}$ rational or irrational? Explain.

◆ SCENARIO

The astrophysicists at the European Southern Observatory measure the distance to a star in kilometres and the diameter of an atom in metres. One number is staggeringly large, the other unimaginably small, yet both are written neatly using the same idea: powers of ten. This unit is about taming the very large, the very small, and the numbers that refuse to be written as fractions.

UNIT 1 • TOPIC 1.1

Irrational numbers

Learn

A **rational number** can be written as a fraction of two integers, such as $\frac{3}{4}$, 7 or 0.5. Every rational number, written as a decimal, either ends or repeats. Some numbers do neither. $\sqrt{2} = 1.41421356\dots$ goes on for ever without repeating, and can never be written exactly as a fraction. Numbers like this are **irrational**. So are $\sqrt{3}$, $\sqrt{5}$ and the famous constant π .

■ WORKED EXAMPLE 1.1

Say whether each number is rational or irrational: a. $\sqrt{16}$ b. $\sqrt{2}$ c. 0.75

- a. $\sqrt{16} = 4$, which is an integer: rational
- b. $\sqrt{2} = 1.41421356\dots$ never ends or repeats: irrational
- c. $0.75 = \frac{3}{4}$, a fraction: rational

Why: A number is rational exactly when it can be written as a fraction of two integers, which means its decimal ends or repeats.

• PRACTISE 1.1

- 1 Say whether each number is rational or irrational: a. $\sqrt{36}$ b. $\sqrt{10}$ c. $0.\dot{3}$
a. $\sqrt{7}$ e. $2/5$.
- 2 Write down an irrational number between 2 and 3.
- 3 a. Write 0.4 as a fraction. b. Is 0.4 rational or irrational?
- 4 Between which two consecutive whole numbers does $\sqrt{20}$ lie?
- 5 True or false: "Every square root is irrational." Give a reason and an example.

∴ THINK LIKE A MATHEMATICIAN

The ancient Greeks were shocked to discover that $\sqrt{2}$ is irrational. Legend says Hippasus was drowned for revealing it. A proof goes like this: suppose $\sqrt{2} = a/b$ in lowest terms, and chase the consequences until both a and b must be even, which is a contradiction. Why does "both must be even" break the assumption "in lowest terms"? Discuss.

■ KEYWORDS

- **rational number:** a number that can be written as a fraction of integers.
- **irrational number:** a number whose decimal never ends and never repeats.

UNIT 1 • TOPIC 1.2

Standard form

Learn

Standard form writes a number as $A \times 10^n$, where A is at least 1 but less than 10, and n is an integer. It is how scientists handle the very large and the very small without drowning in zeros. So $4500 = 4.5 \times 10^3$, and the tiny $0.00072 = 7.2 \times 10^{-4}$.

■ WORKED EXAMPLE 1.2

Work out $(3.2 \times 10^4) \times (2 \times 10^3)$, giving your answer in standard form.

multiply the front numbers: $3.2 \times 2 = 6.4$
add the indices: $10^4 \times 10^3 = 10^7$
so the answer is 6.4×10^7

Why: The front numbers multiply and the powers of ten combine by adding indices, exactly as the laws of indices promise.

● PRACTISE 1.2

- 1 Write in standard form: a. 72 000 b. 0.00045 c. 9 300 000.
- 2 Write as an ordinary number: a. 5.6×10^{-8} b. 3.4×10^5 .
- 3 Work out, in standard form: a. $(4 \times 10^5) \times (2 \times 10^2)$ b. $(9 \times 10^6) \div (3 \times 10^3)$.
- 4 Work out $3 \times 10^5 + 4 \times 10^4$, giving your answer in standard form.
- 5 The distance from Earth to the Sun is about 1.5×10^8 km. Write this as an ordinary number.

UNIT 1 • TOPIC 1.3

Indices

Learn

You know that $5^0 = 1$ and that indices add and subtract. Now meet two more ideas. A **negative index** means a reciprocal: $2^{-3} = 1/2^3 = 1/8$. A **fractional index** means a root: $9^{(1/2)} = \sqrt{9} = 3$, and $8^{(2/3)} = (\sqrt[3]{8})^2 = 4$.

■ WORKED EXAMPLE 1.3

Work out: a. $4^{(3/2)}$ b. 5^{-2} c. 8^0

a. $4^{(3/2)} = (\sqrt{4})^3 = 2^3 = 8$
b. $5^{-2} = 1/5^2 = 1/25$
c. $8^0 = 1$

Why: A fractional index splits into a root and a power; a negative index flips to a reciprocal; anything to the power zero is one.

- **PRACTISE 1.3**

- 1 Work out: a. $25^{(1/2)}$ b. $27^{(1/3)}$ c. $16^{(1/2)}$.
- 2 Work out: a. 3^{-2} b. 10^{-3} c. 2^{-4} .
- 3 Work out $8^{(2/3)}$ and $4^{(3/2)}$.
- 4 Write as a power of 2: a. $1/8$ b. 32.
- 5 Work out $(9^{(1/2)})^2$ and explain your answer.

- **KEYWORDS**

- **standard form:** writing a number as $A \times 10^n$, with $1 \leq A < 10$.
- **negative index:** a power meaning a reciprocal, as in $2^{-3} = 1/8$.
- **fractional index:** a power meaning a root, as in $9^{(1/2)} = 3$.

UNIT 1 • FINAL PROJECT

The scale of everything

★ FINAL PROJECT

The scale of everything

Make a poster showing the scale of the universe, from the diameter of an atom to the distance to a far galaxy, with every measurement written in standard form. Your poster should make the enormous range of sizes feel real.

What to hand in

- At least six real measurements, each written in standard form.
- Each measurement checked against a real source.
- A short note on why standard form is the right tool for the job.

How it will be judged

Criterion	Weight	--- ---	Correct standard form throughout	40%	Range and accuracy of measurements	30%
Quality of presentation	20%		The explanatory note	10%		

UNIT 1 · CHECK YOUR PROGRESS

- 1 Say whether each is rational or irrational: a. $\sqrt{49}$ b. $\sqrt{5}$ c. 0.6.
- 2 Write in standard form: a. 580 000 b. 0.000031.
- 3 Write 2.4×10^6 as an ordinary number.
- 4 Work out $(5 \times 10^4) \times (3 \times 10^3)$, in standard form.
- 5 Work out: a. $36^{1/2}$ b. 4^{-2} c. 6^0 .
- 6 Work out $27^{2/3}$.
- 7 Write $1/16$ as a power of 2.
- 8 Between which two consecutive whole numbers does $\sqrt{50}$ lie?

UNIT 1 · EVALUATION

Be honest. For each outcome, colour the circle that matches how you feel: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
distinguish rational and irrational numbers	☐	☐	☐
write numbers in standard form	☐	☐	☐
calculate in standard form	☐	☐	☐
use zero, negative and fractional indices	☐	☐	☐

UNIT 1 · WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- | | |
|---|---|
| ☐ Tell rational from irrational numbers. | ☐ Use zero, negative and fractional indices. |
| ☐ Write and calculate in standard form. | ☐ Estimate with roots and powers. |

■ UNIT 1 ROUND-UP

rational number · irrational number · standard form · negative index · fractional index

GLOSSARY

Unit 1

The mathematical words this unit has taught you, gathered in one place.

common factor: a number that is a factor of two or more numbers.

common multiple: a number that is a multiple of two or more numbers.

consecutive: following one after another in order.

cube number: a number made by multiplying an integer by itself twice, such as $27 = 3^3$.

cube root: the number that cubes to give the value under the root, as in $\sqrt[3]{27} = 3$.

digit: any of the symbols 0 to 9 used to write numbers.

divisible: able to be divided exactly, with no remainder.

factor: a number that divides another exactly.

highest common factor (HCF): the largest common factor of two or more numbers.

index: the small raised number showing a power, as in 5^2 .

integer: a positive or negative whole number, or zero.

inverse: the number that undoes another; the inverse of 4 is -4 .

inverse operation: an operation that reverses another; subtraction is the inverse operation of addition.

lowest common multiple (LCM): the smallest common multiple of two or more numbers.

multiple: the result of multiplying a number by a positive integer.

number line: a line on which numbers are placed in order.

product: the result of a multiplication.

quotient: the result of a division.

square number: a number made by multiplying an integer by itself, such as $36 = 6^2$.

square root: the number that squares to give the value under the root, as in $\sqrt{36} = 6$.

test for divisibility: a quick rule that tells you whether one number divides another without doing the division.

SOURCES & REFERENCES

Where everything came from

Every worked-example answer and exercise value in this book was checked by a computer before publication.

Factual notes can be checked here. Links were live as of July 2026.

§ UNIT 1 • NUMBER AND CALCULATION

- 1 **Rational and irrational numbers.** The distinction between numbers that can be written as a fraction and those that cannot. <https://en.wikipedia.org/wiki/Irrational_number>
- 2 **The irrationality of root 2.** The classic proof that the square root of two cannot be written as a fraction. <https://en.wikipedia.org/wiki/Square_root_of_2>
- 3 **Standard form and indices.** Scientific notation and the laws of exponents, including zero, negative and fractional powers. <https://en.wikipedia.org/wiki/Scientific_notation> <<https://en.wikipedia.org/wiki/Exponentiation>>

Mathematics that explains itself

This Year 9 Student Book takes you across the whole number line and into the patterns that sit beneath arithmetic. Negative numbers stop being strange, multiples and factors become tools, and powers reveal their roots. Every idea is introduced with a reason, every example is worked in full, and every fact can be checked.

Written for Prime School pupils, in clear British English, with original scenarios drawn from life in Portugal.

INTEGERS

ALGEBRA

NUMBER

GEOMETRY

STATISTICS